

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
First , we need to calculate the quantity of heat required to heat water from 30 °C to 60 °C . The formula to calculate heat (Q) is $Q = mc\Delta T$, where m is the mass of water , c is the specific heat capacity , and ΔT is the change in temperature . The mass of 1 liter of water is approximately 1 kg . The change in temperature (ΔT) is $60 - 30 = 30\text{ }^{\circ}\text{C}$.

$$Q_{\text{water}} = 1\text{ kg} * 1\text{ cal/g}^{\circ}\text{C} * 30\text{ }^{\circ}\text{C} = 30\text{ kcal}$$

Next , we calculate the quantity of heat required to heat the 1 kg block of aluminum from 30 °C to 60 °C .

